



Question Set

Question 1

A market-risk team must reduce factor sensitivities using two hedges. Hedge x reduces equity delta by 6 and credit-spread CS01 by 1. Hedge y reduces equity delta by 2 and CS01 by 4. The desk needs at least 120 units of delta reduction and 80 units of CS01 reduction. Costs are 5 and 4 per unit. Formulate and solve the linear program.

Question 2

Derive the dual of the hedge optimization problem in Question 1 and interpret the dual variables.

Question 3

Use complementary slackness to solve the dual from Question 2 and verify strong duality.

Question 4

A VaR-constrained allocation chooses weights w in two strategies. Expected returns are 8 percent and 12 percent. Risk-budget usage per dollar is 2 and 5. Capital usage per dollar is 4 and 3. Maximize expected return subject to risk budget 10, capital budget 12, and long-only weights. Solve graphically by corner checking.

Question 5

Prove that if an LP has a finite optimum over a nonempty bounded polygon, at least one optimum is attained at an extreme point.

Question 6

A hedge LP is infeasible after the desk adds liquidity and notional caps. Give a rigorous diagnostic checklist that a quant interviewer would expect.

Question 7

Formulate an L1 transaction-cost hedge that minimizes absolute residual risk exposure after trading two hedges.

Question 8

What is degeneracy in an LP, and why can it matter in hedge optimization interviews?

Question 9

A portfolio optimization LP has multiple optimal solutions along an edge. How would you choose a practical hedge?

Question 10

Explain why dual prices are local quantities. Give a market-risk example.

Question 11

A desk minimizes funding cost

$$c^\top x$$

subject to

$$Ax \geq b$$

Explain the economic meaning of reduced cost for an unused funding source.

Question 12

How can a linear stress-test allocation problem become misleading even if the LP is solved exactly?